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LOS LAB 6 Theory

1) Caesar cipher

Caesar send the ciphertext EVIRE. Antony does not know the key. He tries all possible keys to find original message

Cryptanalysis model: Ciphertext-only attack

a) what meaningful plaintexts does Antony obtain

The encryption formula

$$C = (P + K) \bmod 26$$

To decrypt, we reverse the shift

$$P = (C - K) \bmod 26$$

Converting :- for this question, Antony knows the ciphertext EVIRE, but he does not know K

Step 2 Convert the ciphertext into numbers

Ciphertext:

EVIRE

Ciphertext letter: E V I R E

Numerical val: 4 21 8 17 4

C = (4, 21, 8, 17, 4)

Try key K = 0

$$P = (C - K) \bmod 26$$

Take K = 0, we subtract zero for every letter

For E:

$$P = (4 - 0) \bmod 26 = 4 = E$$

For V: $P = (21 - 0) \bmod 26 = 21 = V$

For I: $P = (8 - 0) \bmod 26 = 8 = I$

for R: $P = (17 - 0) \bmod 26 = 17 = R$

For E: $P = (4 - 0) \bmod 26 = 4 = E$

∴ output: EVIRE

Try key = 1

for E: $P = (4-1) \bmod 26 = 3 \bmod 26 = 3 = D$

for V: $P = (21-1) \bmod 26 = 20 = U$

for I: $P = (8-1) \bmod 26 = 7 = H$

R: $P = (17-1) \bmod 26 = 16 = Q$

E: $P = (4-1) \bmod 26 = 3 = D$

∴ output: DUHQD

~~continue k=4~~

Now try key = 4

E:

$P_1 = (4-4) \bmod 26 = 0 = A$

V: $P_2 = (21-4) \bmod 26 = 17 = R$

I: $P_3 = (8-4) \bmod 26 = 4 = E$.

R: $P_4 = (17-4) \bmod 26 = 13 = N$

E: $P_5 = (4-4) \bmod 26 = 0 = A$.

Output: ARENA

Nw try key k = 13

E: $P_1 = (4-13) \bmod 26$

$= -9 \bmod 26$

Add 26 to bring the answer into range 0-25

$= -9 + 26 = 17 = R$

V: $P_2 = (21-13) \bmod 26 = 8 = I$

I: $P_3 = (8-13) \bmod 26$

$= -5 + 26 = 21 = V$

R: $P_4 = (17-13) \bmod 26 = 4 \bmod 26 = 4 = E$

E: $P_5 = (4-13) \bmod 26$

$= -9 + 26 = 17 = R$

output: RIVER

This is so meaningful.

Stp: A Caesar cipher has 26 possible keys.

$k = 0, 1, 2, 3 \ldots 25$

The two intended meaningful results in this

4: AREMA — Refer to the Colosseum

13 RIVER — Refer to the Tiber RIVER

b) When should Antony meet Caesar?

We have two clues

- ARENA refers to the Colosseum.
- RIVER refers to the Tiber RIVER

The question says Caesar is choosing b/w the Tiber River and the Colosseum. The ciphertext gives two possible meaningful messages when different key are tried

Therefore, the ciphertext alone not uniquely tells Antony which location Caesar intended. He needs additional information, such as a previously agreed key or another clue, to decide conclusively.

c) What do we learn from a ciphertext-only attack

The lesson is that an attacker can try every possible Caesar key without knowing the original key.

This is called a brute-force attack

since there are only 26 keys, the attack can test all of the quickly. But the attacker may obtain more than one plausible plaintext

2) Affine Cipher - Finding the Encryption key

Cryptanalysis model: Chosen-plaintext attack

a) Convert the plaintext ad ciphertext letters into numerical values

Plaintext:

HAHAHA

H, A, H, A, H, A

H=7, A=0

Therefore P = (7, 0, 7, 0, 7, 0)

Ciphertext

NONONO

N, O, N, O, N, O

using the letter table

N=13, O=14

C = 13, 14, 13, 14, 13, 14

Plaintext	P	Ciphertext	C
H	7	N	13
A	0	O	14
Repeat order	:	:	:

b) Form the equation modulo 26

$E(u) = (au+b) \bmod 26$

Use the first pair, H → N

H = 7, N = 13

Substitute u = 7 & E(u) = 13 into the formula

$13 = (7a+b) \bmod 26$

$7a + b \equiv 13 \pmod{26}$

first eq

Use the second pair A → O

A = 0, O = 14

u = 0

$14 = (a \times 0 + b) \bmod 26$

$14 = (0+b) \bmod 26$

$b \equiv 14 \pmod{26}$

Second eq

c) Determine a & b

b=1

$b \equiv 14 \pmod{26}$

$b = 14$

Substitute $b = 14$

first eq: $7a + b = 13 \pmod{26}$

Replace b with 14

$7a + 14 \equiv 13 \pmod{26}$

Subtract 14 from both sides

$7a = 13 - 14 \pmod{26}$

$7a \equiv -1 \pmod{26}$

since add 26 does not change the mod'd

$-1 + 26 = 25$

$7a \equiv 25 \pmod{26}$

Multiplicative inverse of 7

15

$7 \times 15 = 105$

$105 = 26 \times 4 + 1$

$7^{-1} \equiv 15 \pmod{26}$

$7a \equiv 25 \pmod{25}$

$15 \times 7a \equiv 15 \times 25 \pmod{26}$

$105a \equiv 375 \pmod{26}$

since $105 = 1 \pmod{26}$

$a \equiv 375 \pmod{26}$

Now divide 375 by 26

$375 = 26 \times 14 + 11$

$a = 11$

$a = 11$ & $b = 14$

d) Encrypt: first

$E(x) = (ax + b) \bmod 26$

$a = 11$ & $b = 14$

$E(x) = (11x + 14) \bmod 26$

verify the answer

H, n = 7

$E(7) = (11 \times 7 + 14) \bmod 26$

$= (77 + 14) \bmod 26$

$= 91 \bmod 26$

$91 = 26 \times 3 + 13$

for A, n = 0;

$E(0) = (11 \times 0 + 14) \bmod 26$

$= 14 \bmod 26 = 14 = O$

Thus H becomes N & A becomes O. Repeating this gives NONONO, as required

3) Affine Cipher – Decryption of CRWW2

Cryptanalysis model : known plaintext attack

a) Determine a & b

write the known plaintext & ciphertext

Known plaintext

HA

Ciphertext

CRWW2

Plaintext	Plaintext value	Ciphertext	Ciphertext value
H	7	C	2
A	0	R	17

H → C

A → R

write the Affine encryption formula

$E(n) = (an + b) \bmod 26$

Using H → C

H → 7, C → 2

$2 = (7a + b) \bmod 26$

$7a + b \equiv 2 \pmod{26}$

Use A→R

A→0, R→17

$17 = (a \times 0 + b) \bmod 26$

$17 = b \bmod 26$

$b = 17$

Substitute $b = 17$

$7a + b = 2 \pmod{26}$

Substituted

$7a + 17 \equiv 2 \pmod{26}$

Subtract 17 from both sides

$7a \equiv 2 - 17 \pmod{26}$

$7a \equiv -15 \pmod{26}$

$-15 + 26 = 11$

$7a \equiv 11 \pmod{26}$

We need a number that makes the product with 7 like mod...

$7 \times 15 = 105$

$105 = 26 \times 4 + 1$

$\therefore$

$7^{-1} \equiv 15 \pmod{26}$

calculate a

Multiply both sides by 15

$15 \times 7a \equiv 15 \times 11 \pmod{26}$

$105a \equiv 165 \pmod{26}$

Because $105 \equiv 1 \pmod{26}$

$a = 165 \pmod{26}$

Divide 165 by 26

$165 = 26 \times 6 + 9$

The remainder is 9:

$a = 9$

$a = 9, b = 17$

b) Write the encryption function

$E(x) = (ax + b) \bmod 26$

substitute the key value

$a = 9, b = 17$

$E(x) = (9x + 17) \bmod 26$

c) Find the inverse of a & with the decryption function

$a = 9$ modulo 26

$26 = 9 \times 2 + 8$

$9 = 8 \times 1 + 1$

Rearrange the second equation

$1 = 9 - 8$

from the first equation

$8 = 26 - 2 \times 9$

Substitute

$1 = 9 - (26 - 2 \times 9)$

$1 = 3 \times 9 - 26$

$3 \times 9 \equiv 1 \pmod{26}$

So the multiplicative inverse

$9^{-1} = 3 \pmod{26}$

so the multiplicative inverse is

$9^{-1} = 3 \pmod{26}$

$D(y) = 3(y - 17) \bmod 26$

$Dy = (3y - 51) \bmod 26$

since $-51 + 52 = 1$ we have $-5 \equiv 1 \pmod{26}$

$D(y) = (3y + 1) \bmod 26$

d) Decrypt the complete ciphertext

Ciphertext: CRWWZ

$C = (2, 17, 22, 22, 25)$

Use the decryption formula

$D(y) = (3y + 1) \bmod 26$

$C = 2$

$D(2) = (3 \times 2 + 1) \bmod 26$

$7 = 4$

for $R = 17$;

$D(17) = (3 \times 17 + 1) \bmod 26$

$= 52 \bmod 26 = 0 = A$

for the first u = 22

$D(22) = (3 \times 22 + 1) \bmod 26$

$= 67 \bmod 26 = 15 = P$

$D(22) = 67 \bmod 26 = 15 = P$

for z = 25:

$D(25) = (3 \times 25 + 1) \bmod 26$

$= 76 \bmod 26 = 24 = Y$

HAPPY

4) On - Tim Pad

a) Convert the plaintext & key into numbers

The Plaintext is RIVER

$P = (17, 8, 21, 4, 12)$

The key is NAXTP

$K = (13, 16, 23, 19, 15)$

b) Encrypt the message

$C_i = (P_i + K_i) \bmod 26$

R = 17 & N = 13

$C_1 = (17 + 13) \bmod 26$

$= 30 - 26 = 4 = E$

for I = 8 & A = 16

$C_2 = (8 + 16) \bmod 26 = 24 = Y$

for V = 21 & X = 23:

$C_3 = (21 + 23) \bmod 26$

$44 - 26 = 18 = S$

for E = 4 & T = 19

$C_4 = (4 + 19) \bmod 26 = 23 = X$

R = 17 & P = 15

$C_5 = (17 + 15) \bmod 26$

32 - 26 = 6 = G

The ciphertext is:

E Y S X G

c) Decrypt the ciphertext

$P_i = (C_i - K_i) \bmod 26$

Ciphertext: E Y S X G

for E = 4 & N = 13

$P_1 = (4 - 13) \bmod 26$

$= -9 + 26 = 17 = R$

Y = 24 & A = 16

$P_2 = (24 - 16) \bmod 26 = 8 = I$

for S = 18 & X = 23

$P_3 = (18 - 23) \bmod 26$

$= (203 - 19) \bmod 26 \bmod = 4 = E$

for G = 6 & P = 15

$P_3 = (6 - 13) \bmod 26$

$= -7 + 26 = 17 = R$

RIVER

d) Why must the key be random, equal in length, & used only once

1) Random: The key must be unpredictable. If it follows a predictable pattern, an attacker may be able to guess it

2) Equal in length: Each plaintext letter needs a corresponding key letter. Thus RIVER & NEXTP NEXTP both five letter

3. Used only space: Reusing the same key for different messages can reveal relationship b/w the messages & weaken security

4. A One time pad provides perfect secrecy when the key is truly random, secret, equal in length to the plaintext and used only once.